

Iliyaan Bhulani

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EDUCATION

Georgia Institute of Technology

Bachelor's of Science in Computer Science

Atlanta, GA

Expected Graduation: May 2028

- **Concentrations:** Artificial Intelligence and Cybersecurity
- **Awards:** Dean's List, Faculty Honors, EFAS Scholarship, GHP Scholar
- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Artificial Intelligence, Machine Learning, Robotics and Perception

EXPERIENCE

Murty Lab — Georgia Institute of Technology

Undergraduate Researcher

Atlanta, GA

September 2025 – Present

- Developing deep neural network-based voxelwise encoding models to predict neural activity in macaque neurons and human fMRI voxels, studying visual information processing across brain regions including V1, FFA, EBA, and PPA.
- Mapped visual receptive fields from Bashivan macaque neural recordings to characterize spatial tuning properties across the visual hierarchy.
- Investigated sampling efficiency and generalization in encoding models, examining how distributional distance between training and test stimuli shapes model performance.
- Examined intracranial single-neuron data via neuron-voxel encoding models using NSD.

Georgia Institute of Technology College of Computing

Teaching Assistant

Atlanta, GA

August 2024 – May 2025

- Assisting **500+** students with homework, logistical, and conceptual materials related to the course, as well as exam preparation through recitations and office hours.
- Elaborating on advanced topics, such as Data Structures and Object-Oriented Programming in Python, raising student scores by an average of **6%**.

Microsoft (Realms Project)

Intern

Atlanta, GA

January 2023 – February 2024

- Developed a map for the popular video game “Minecraft” as a realms map-maker. Played by **50,000+** players.
- Built **data packs** and **mod packs** using Java, JSON, and Blender, improving game features and user experience.

PROJECTS

Cognitive Load | *Python, Machine Learning, React.js, EEG/PPG/rPPG Analysis*

August 2025 – Present

- Contributing to machine learning models predicting cognitive load (stress, relaxed, neutral) using EEG, PPG, and video-based rPPG signals.
- Exploring video-based analysis methods to extract heart rate variability (HRV) metrics for clinical applications.
- Supporting clinical trial outreach through IRB submission and mentor collaboration.

HarmonyHub | *Python, Machine Learning, Django, React*

August 2024 – September 2024

- Designed and implemented a Spotify Wrapped-inspired web application to generate personalized listening statistics using machine learning models, leveraging Django, Python, and React.
- Created interactive dashboards with tailored insights, including top songs, genres, and listening trends, to enhance user engagement with detailed visualizations.

Brain Tumor Classifier | *Python, TensorFlow, Keras, Machine Learning*

August 2025 – September 2025

- Developed a convolutional neural network using MobileNetV2 to classify human brain MRI images into glioma, meningioma, pituitary tumor, or no tumor, leveraging transfer learning for improved accuracy.
- Implemented Grad-CAM visualizations to highlight regions of interest on MRI scans, providing interpretable insights into model predictions.

LEADERSHIP

Evergreen Center

Youth Team Lead

Tucker, GA

August 2025 – Present

- Led a team of **30+** youth volunteers to develop and execute programs that enhanced engagement and support for senior center participants.
- Coordinated events and initiatives that fostered intergenerational connections, promoted wellness, and strengthened community involvement.

TECHNICAL SKILLS

Languages: Python (Intermediate), Java (Intermediate), JavaScript/HTML/CSS (Intermediate), C++ (Familiar), SQL (Familiar)

Frameworks & Libraries: Node.js (Intermediate), React.js (Familiar), Next.js (Familiar), Django (Familiar), Pandas (Familiar), scikit-learn, NumPy

Machine Learning & Data:: EEG/PPG/rPPG signal processing, Heart Rate Variability (HRV) analysis, Predictive modeling, Real-time data visualization

Technologies: Linux, Git/GitHub, Docker, Puppeteer, Chromium, AWS, NVIDIA LLM Models, Trello